

NEXHIRE

Enterprise Product, Business and Requirements Specification

Vision, BRD, Scope, Requirements, Rules, Stakeholders and Product Governance

Volume 1 of 4 | Version 3.0 | Team ALPHA | 5 August 2026

Enterprise Second-Brain Edition - Controlled Source of Truth

Document Control and Version History

Field	Value
Product	NexHire Smart Recruitment Matching Platform
Document Set	Enterprise Documentation and Second Brain v3.0
Owner	Kogul / Team ALPHA
Classification	Project Internal / Academic Submission / Client-Ready
Canonical Product Baseline	Second Brain Final Pack v2.0
Implementation Overlay	Master Documentation and Team Handover Pack
Review Rule	Changes require an approved decision record and traceability update

Version	Date	Status	Summary
1.0	4 Aug 2026	Historical	Original source documentation baseline
2.0	5 Aug 2026	Validated Source	Second Brain canonical pack and validated prototype
3.0	5 Aug 2026	Current Controlled Edition	Unified enterprise documentation with conflicts resolved through phase boundaries

Authoritative interpretation rule

Core Release v1.0 follows the trust-first, explainable matching, duplicate-safe application and consent-gated contact scope of the validated Second Brain pack. Interview, Offer, Talent Pool and Complaint capabilities from the implementation overlay are retained as Phase 2 extensions, not silently mixed into Core Release v1.0.

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1. Executive Summary

NexHire is a trust-first smart recruitment matching platform designed to reduce low-quality vacancy exposure, opaque candidate ranking, uncontrolled personal-data disclosure and manual recruitment friction. The product combines verified employer identity, moderated vacancy publication, structured candidate evidence, deterministic and explainable matching, duplicate-safe applications, auditable state transitions and candidate-controlled contact consent.

The documentation defines a controlled Core Release v1.0 and a separately governed Phase 2 roadmap. Core Release v1.0 ends with candidate consent to an employer contact request. Interview scheduling, offer management, talent pools and complaint case management are retained as enterprise extensions and require approved change records before activation. This separation resolves the scope conflict found across the two source packs while preserving all documented product ideas.

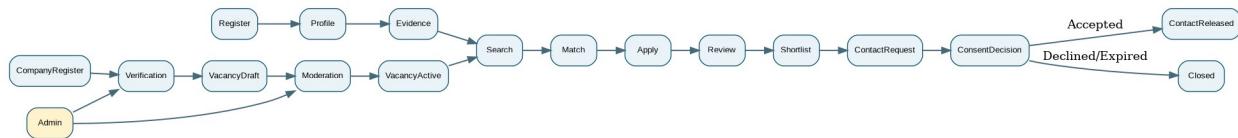


Figure 1 - End-to-end product workflow and trust/consent boundaries

Canonical matching decision

Core Release v1.0 uses RM-1.0: Required Skills 55%, Experience 20%, Education 15%, Location 10%. Preferred skills do not score in Core Release. Any alternative weighting is a future version and requires rule versioning, approval, golden fixtures and impact analysis.

2. Project Overview and Background

2.1 Product Vision

Enable trustworthy, evidence-based recruitment decisions while preserving candidate agency and providing employers with a clear, explainable and operationally efficient applicant workflow.

2.2 Mission

Create a recruitment platform where employer trust is verified, candidate evidence is structured and protected, matching decisions can be explained and replayed, applications remain transactionally correct, and personal contact information is disclosed only through explicit job-specific consent.

2.3 Strategic Product Principles

- Trust before reach: unverified or unmoderated vacancies are not promoted as trusted opportunities.
- Evidence before inference: candidate skills and profile data must be structured and attributable.
- Explainability before automation: scores show components, gaps and rule version.
- Consent before disclosure: candidate phone/email remain masked until an accepted contact request.
- Database truth before dashboard decoration: reports reconcile with authoritative records.
- State machines before ad-hoc status text: transitions are explicit, validated and auditable.

3. Business Problem and Existing System

3.1 Existing Recruitment Environment

Current recruitment ecosystems commonly combine public job boards, social-professional networks, applicant tracking systems and employer career sites. They provide scale but frequently create fragmented trust signals, inconsistent skills taxonomies, opaque ranking, duplicate application behaviour, uncontrolled CV distribution and weak candidate visibility into application decisions.

3.2 Existing System Limitations

Limitation	Business Impact	NexHire Response
Unverified or weakly verified employers	Fraud risk and reduced candidate confidence	Manual verification, evidence history and trusted public predicate
Keyword-driven matching	False positives and missed qualified candidates	Canonical skills and deterministic components
Opaque ranking	Low trust and difficult dispute resolution	Symmetric explanations and replay receipt
Duplicate applications	Data inconsistency and recruiter noise	Unique constraint plus idempotent command
Uncontrolled CV/contact exposure	Privacy and safety risk	Private streaming, masking and consent release
Manual status handling	Inconsistent candidate experience	Allowed transitions, history, notifications and audit
Unreconciled dashboards	Misleading management decisions	Database-truth reports and formula-safe exports

3.3 Consolidated Business Problem

Job seekers need trustworthy vacancies, clear fit explanations and control over personal data. Employers need verified operating identities, reliable candidate evidence, stable ranking and efficient review. Administrators need transparent governance without becoming hidden score operators. The platform must satisfy these needs through consistent domain rules rather than presentation-only controls.

4. Proposed Solution and Objectives

4.1 Proposed Solution

NexHire is implemented as a modular web platform with role-specific workspaces for Job Seekers, Employers and Administrators, supported by a background worker. The solution manages identity, structured evidence, controlled skills, private documents, company trust, vacancy lifecycle, deterministic matching, applications, consent contact, notification delivery, reporting and governance.

4.2 Project Objectives

1. Deliver a secure role-based recruitment platform.
2. Increase candidate trust through verified employers and vacancy moderation.
3. Provide transparent, deterministic matching and skill-gap explanations.
4. Prevent duplicate application effects under retries and concurrency.
5. Protect candidate CV and contact data through resource authorization and explicit consent.
6. Support auditable recruitment and administration decisions.
7. Provide a maintainable architecture suitable for academic demonstration and enterprise evolution.

4.3 Success Measures

Measure	Target / Acceptance Direction
Trusted vacancy integrity	Only verified active companies and active non-expired moderated vacancies appear publicly
Match reproducibility	Golden fixtures produce identical component and total scores for a rule version

Measure	Target / Acceptance Direction
Duplicate prevention	Concurrent/retried apply attempts produce one application
Contact privacy	No contact detail returned before accepted valid request
Auditability	All trust, moderation, consent and account-state decisions have immutable audit events
Usability	Critical flows keyboard-operable with actionable error and confirmation states
Reliability	No partial business effect between application/status commit and notification outbox

5. Scope, Out of Scope, Assumptions and Constraints

5.1 Core Release v1.0 In Scope

- Registration, login, refresh, logout, role and account-state enforcement.
- Structured candidate evidence and readiness.
- Canonical skills, aliases and candidate evidence.
- Private upload, scan, controlled streaming and deterministic template rendering.
- Company profile, evidence and verification lifecycle.
- Draft, moderation, activation, closure and expiry.
- Trusted public vacancy search and candidate projections.
- RM-1.0 deterministic scoring, gaps, receipts and replay.
- Idempotent duplicate-safe apply, history and withdrawal.
- Stable ranking, evidence review and protected CV access.
- Allowed transition matrix, history and notifications.
- One pending request, candidate decision, conditional detail release.
- Owner-only in-app inbox and durable outbox delivery.
- Preferences and controlled deletion requests.
- Truth-based summaries, exports and append-only audit.
- Verification, users, moderation, skill governance and locked settings.

5.2 Core Release v1.0 Out of Scope

- Interview scheduling and interview evaluation
- Offer lifecycle and onboarding
- Talent pool management
- Real-time chat and video interview hosting
- Autonomous AI/ML CV inference or final hiring decision
- Payments and subscription billing
- Background-check provider integration
- External email/SMS delivery unless separately approved
- Native mobile applications

5.3 Phase 2 Planned Extensions

Extension	Purpose	Activation Condition
Interview Management	Schedule, reschedule, cancel and evaluate shortlisted candidates	Approved CR with APIs, entities, security and test traceability
Offer Management	Draft, send, accept, reject, expire and withdraw offers	Approved lifecycle and legal/retention rules
Talent Pool	Retain consented candidate opportunities	Consent, retention and ownership policy

Extension	Purpose	Activation Condition
	for future vacancies	approved
Complaint Case Management	Structured user complaints beyond vacancy reporting	Case taxonomy, SLA, privacy and admin workflow approved
RM-2.0 Matching	Optional preferred-skill and evidence factors	Fairness review, owner approval, versioned golden fixtures

5.4 Assumptions

- The initial product is a web platform.
- Users have unique email identities.
- Employer verification is performed by authorized administrators.
- The skill dictionary is governed centrally.
- The initial workload fits a modular-monolith deployment while preserving horizontal API scale.
- The exact runtime version is a versioned technical baseline, not an immutable product rule.

5.5 Constraints and Dependencies

- University project delivery timeline and team capacity.
- SQL Server-compatible relational persistence.
- Private object storage for CV and verification evidence.
- Current repository and OpenAPI are required for exact implementation-name reconciliation.
- Retention, jurisdiction and production hosting decisions require owner approval.

6. Stakeholders and User Roles

Stakeholder / Role	Primary Needs	Accountability
Job Seeker	Maintains evidence, searches trusted vacancies, views match explanations, applies, tracks applications, decides contact consent.	Uses only authorized resources and follows documented states.
Employer	Maintains company evidence, submits verification, manages vacancies, reviews ranked applicants, updates application status, requests contact.	Uses only authorized resources and follows documented states.
Administrator	Operates trust, verification, moderation, account administration, skill governance, audit review and locked settings.	Uses only authorized resources and follows documented states.
System Worker	Executes expiry, outbox delivery, scan lifecycle, notification and state-guard jobs without public ingress.	Uses only authorized resources and follows documented states.
Project Owner / Product Manager	Scope, priorities, release decisions and value realization	Approves product and matching changes.
Development Team	Stable contracts, module ownership and test evidence	Implements against the controlled documentation.
University Evaluators / Client	Traceable requirements, architecture and demonstration evidence	Reviews fitness, completeness and quality.

7. Product Constitution and Release Boundaries

Non-negotiable product identity

NexHire must remain a verified-employer, evidence-based, explainable and consent-controlled recruitment platform. Technology and code organization may evolve; trust predicates, role separation, score explainability, application integrity and contact-consent meaning may not be silently removed.

Constitution Rule	Required Interpretation
Canonical product terms	Use Job Seeker, Employer, Administrator and System Worker in documentation; code aliases must be mapped.
Core lifecycle	Registration -> Evidence -> Trusted Vacancy -> Match -> Apply -> Review -> Status -> Contact Request -> Candidate Decision.
Matching	Rule-versioned and deterministic. No hidden or individual score override.
Privacy	Contact and private documents are resource-authorized and masked by default.
Scope change	Phase 2 features require change records and traceability before becoming Core.
Source hierarchy	Product constitution and approved decisions -> business rules -> architecture/API/data -> implementation overlay -> prototype.
Conflict handling	Do not blend conflicting rules; record and approve the decision.

8. Functional Requirements

ID	Requirement
FR-01	Account registration for Job Seeker and Employer with notice acknowledgement.
FR-02	Secure authentication, rotating refresh session and role workspace routing.
FR-03	Structured candidate profile and readiness calculation.
FR-04	Controlled skill passport using canonical skills and aliases.
FR-05	Protected CV upload, scan, private retrieval and deletion.
FR-06	Candidate-owned CV Studio with schema validation and approved PDF rendering.
FR-07	Search and filter only trusted, active, non-expired vacancies.
FR-08	Explainable deterministic matching with gaps and replay receipt.
FR-09	Duplicate-safe, idempotent job application.
FR-10	Application history, detail and allowed withdrawal.
FR-11	Candidate decision on employer contact request.
FR-12	Owner-only in-app notifications.
FR-13	Candidate privacy preferences and controlled deletion request.
FR-14	Company profile and manual verification submission.
FR-15	Vacancy draft, edit, activate, close and expire lifecycle.
FR-16	Ranked applicant review and protected evidence access.
FR-17	Employer-owned application status transitions.
FR-18	Consent-gated employer contact request and accepted detail release.
FR-19	Employer summary and formula-safe export.
FR-20	Admin verification decisions.
FR-21	Admin account management and dashboard.
FR-22	Vacancy reporting and moderation.
FR-23	Canonical skill dictionary governance.
FR-24	Audit review and locked platform settings.
FR-25	Automatic lifecycle expiry and durable event processing.

8.1 Functional Acceptance Pattern

Every functional requirement is accepted only when its business rule, use case, API contract, database constraint or state policy, authorization requirement and test evidence are traceable. A visual prototype alone is not implementation proof.

9. Non-Functional Requirements

ID	Requirement
NFR-SEC-01	All protected operations use authenticated HTTPS and server-side role/resource authorization.
NFR-SEC-02	Passwords are identity-managed and never stored or logged in plaintext.
NFR-SEC-03	Refresh tokens are rotated, hashed at rest, revocable and session-bound.
NFR-SEC-04	CV and verification evidence remain private and are streamed only after policy checks.
NFR-SEC-05	Uploads are type/size validated, quarantined and malware-scanned before availability.
NFR-PERF-01	Paged read APIs target p95 under 500 ms at the approved MVP workload, excluding file streaming.
NFR-PERF-02	Match calculation is deterministic and targets p95 under 300 ms for normal skill sets.
NFR-REL-01	Commands are idempotent where retries can duplicate business effects.
NFR-REL-02	Optimistic concurrency protects mutable aggregate state.
NFR-REL-03	Outbox delivery prevents loss between database commit and notification/event publication.
NFR-SCALE-01	API is stateless and horizontally scalable; durable state resides in SQL/private object storage.
NFR-UX-01	All screens support keyboard navigation, visible focus, semantic labels and non-colour status cues.
NFR-OBS-01	Structured logs, metrics and traces carry correlation identifiers without sensitive content.
NFR-AUD-01	Trust, moderation, status, consent and administration decisions create immutable audit events.
NFR-PRIV-01	Candidate contact details remain masked until valid consent is accepted.
NFR-MNT-01	Architecture uses modular boundaries, DTO contracts, validators and testable domain policies.
NFR-BCP-01	Database and private objects have tested backup and restore procedures.
NFR-COMP-01	Retention, deletion and privacy decisions are configurable and documented before production launch.

10. Business Rules and Business Logic

ID	Rule
BR-001	Only verified, active companies with an approved/active vacancy may appear in trusted public search.
BR-002	Required skills drive RM-1.0 skill scoring. Preferred skills are informational in Core Release v1.0 and do not alter score.
BR-003	RM-1.0 weights are Required Skills 55%, Experience 20%, Education 15% and Location 10%.
BR-004	Each candidate-vacancy pair supports one lifetime application in Core Release v1.0.
BR-005	Application creation requires Idempotency-Key and a database uniqueness constraint.
BR-006	Employer application status changes follow the approved transition matrix and create ordered history.
BR-007	Candidate contact details are masked until a job-specific contact request is accepted and context remains valid.
BR-008	A contact request expires after seven days unless decided earlier.
BR-009	Administrators can activate approved rule versions but cannot manually override an individual candidate score.

ID	Rule
BR-010	Every trust, moderation, consent and account-state decision is auditable.
BR-011	A vacancy cannot be re-opened in Core Release v1.0 after closure.
BR-012	Missing profile or evidence produces a safe, explainable result; it must not cause an unhandled server error.
BR-013	Private objects use generated storage identifiers; user-supplied paths are never trusted.
BR-014	Phase 2 Interview, Offer, Talent Pool and Complaint modules are extensions and must not alter Core Release contracts without a change record.

10.1 RM-1.0 Matching Logic

Component	Weight	Calculation Direction
Required Skills	55%	Matched required canonical skills / total required skills. Empty required-skill policy must be explicitly handled.
Experience	20%	Candidate verified experience compared with vacancy minimum. Capped at component maximum.
Education	15%	Approved education compatibility rule; no free-text subjective inference.
Location	10%	Declared candidate preference/location compared with vacancy work mode/location.

The engine stores the rule version, component scores, total score, skill gaps, input snapshot identifiers and calculation timestamp. Stable ranking uses total score followed by a documented deterministic tie-breaker such as application timestamp and application identifier.

10.2 Core State Rules

Aggregate	States
Account	Active, Suspended, Deactivated
Company Verification	Draft, Submitted, UnderReview, Verified, Rejected, Expired
Vacancy	Draft, PendingModeration, Active, Closed, Expired, Rejected
Application	Submitted, UnderReview, Shortlisted, Rejected, Withdrawn
Contact Request	Pending, Accepted, Declined, Expired, Cancelled
Resume	Quarantined, Scanning, Active, Rejected, Deleted
Deletion Request	Submitted, UnderReview, Approved, Rejected, Completed
Vacancy Report	Submitted, UnderReview, Resolved, Rejected

11. User Journeys and Use Cases

11.1 Universal Interaction Contract

Every human-facing use case follows See -> Validate -> Authorize -> Commit -> Confirm. The user sees relevant context, receives client and server validation, passes role/resource authorization, commits atomically, and receives deterministic confirmation or actionable failure.

ID	Use Case
UC01	Register Account
UC02	Login, Role Access and Logout
UC03	Create and Maintain Candidate Profile Evidence
UC04	Manage the Controlled Skill Passport

ID	Use Case
UC05	Upload, Replace or Remove a Protected CV
UC06	Create a Candidate-Owned CV in CV Studio
UC07	Search and Filter Trusted Vacancies
UC08	View Two-Sided Match, Gaps and Replay Receipt
UC09	Submit a Duplicate-Safe Job Application
UC10	Track History or Withdraw an Application
UC11	Accept, Decline or Let a Contact Request Expire
UC12	Read and Manage In-App Notifications
UC13	Manage Candidate Privacy and Deletion Request
UC14	Maintain Company Profile and Submit Verification
UC15	Create, Publish, Edit, Close and Expire a Vacancy
UC16	Review the Ranked Applicant Queue and Protected CV
UC17	Update Employer-Owned Application Status
UC18	Request Candidate Contact and View Accepted Details
UC19	View and Export Basic Employer Reports
UC20	Review and Decide Employer Verification
UC21	Manage Accounts and Basic Admin Dashboard
UC22	Moderate Vacancies and Resolve Candidate Reports
UC23	Govern the Canonical Skill Dictionary
UC24	Review Audit Events and Locked Platform Settings
UC25	Enforce Vacancy and Contact Lifecycle Automatically

11.2 Representative Journey - Job Seeker

8. Register and acknowledge privacy notice.
9. Sign in and enter candidate workspace.
10. Complete structured profile and canonical skill passport.
11. Upload a protected CV or create a CV Studio document.
12. Search only trusted vacancies.
13. Review RM-1.0 explanation and gaps.
14. Apply using a duplicate-safe command.
15. Track ordered application history or withdraw in an allowed state.
16. Receive a job-specific contact request.
17. Accept, decline or allow expiry; contact details are released only after acceptance.

11.3 Representative Journey - Employer

18. Register as Employer.
19. Maintain company profile and submit verification evidence.
20. Wait for administrator decision.
21. Create a Draft vacancy with required and preferred skills.
22. Activate only when trust predicate and validation pass.
23. Review stable ranked applicants and protected evidence.
24. Update application status using allowed transitions.
25. Create a contact request for an eligible applicant.
26. View accepted details only after candidate consent and valid context.
27. Review database-truth summary and safe export.

11.4 Representative Journey - Administrator

28. Review verification evidence and history.

29. Approve or reject with reason and optimistic concurrency.
30. Monitor truth-based dashboard queues.
31. Suspend/reactivate accounts with session revocation.
32. Review vacancy reports and execute moderation.
33. Govern canonical skills and aliases.
34. Review bounded audit events and locked settings.

12. System Workflow, Activity, Sequence and Data Flow

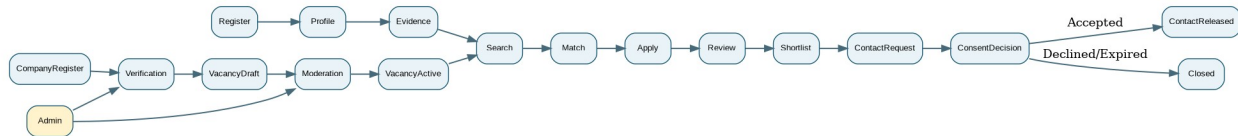


Figure 2 - Activity flow across user roles and trust boundaries

12.1 Application Command Sequence

35. Client submits POST application with JWT and Idempotency-Key.
36. API validates model, account role, profile readiness and vacancy public trust predicate.
37. Application service checks idempotency record and database uniqueness.
38. Matching receipt is obtained or calculated under RM-1.0.
39. Application and initial status history are committed in one transaction.
40. Notification/outbox records are committed in the same transaction.
41. Response returns 201 with application identifier and stable status.
42. Worker delivers durable notification; retries cannot create another application.

12.2 Contact Consent Sequence

43. Employer requests contact for an owned eligible application.
44. System creates one pending request with seven-day expiry.
45. Candidate receives in-app notification.
46. Candidate opens request with company/job context.
47. Candidate accepts or declines using optimistic concurrency.
48. Accepted decision enables context-bound detail endpoint; declined/expired decisions never release details.
49. Decision and access are audited.

12.3 Data Flow Explanation

Public search reads only data that satisfies the complete trust predicate. Candidate evidence flows into matching through normalized snapshots rather than uncontrolled document inference. Application and status commands write transactional records and outbox events. Private documents flow through quarantine and scan before authorized streaming. Contact data remains in the user/profile domain and is conditionally projected only through an accepted request policy.

13. Phase 2 Extension Requirements

The following capabilities are preserved from the implementation overlay but are not part of the canonical Core Release v1.0. They must be added through formal change requests so that existing consent, application and privacy contracts remain stable.

Extension	Minimum Documentation Required Before Build
Interview	Use cases, shortlisted eligibility, schedule state machine, time-zone rules, ownership, notifications, evaluation model, APIs, entities and tests.
Offer	Eligibility, draft/sent/accepted/rejected/expired/withdrawn state machine, compensation validation, candidate ownership, audit and retention.
Talent Pool	Candidate consent, company ownership, source, retention, deletion and future-job contact rules.
Complaints	Complaint categories, evidence, status SLA, privacy, admin resolution and appeal path.
RM-2.0	Fairness impact, approved weights, compatibility, score migration/replay, golden fixtures and rule activation.

14. Conclusion, References and Appendices

14.1 Conclusion

The controlled product definition preserves the strongest verified elements of both source packs without creating a contradictory hybrid. Core Release v1.0 is implementation-ready when exact repository contracts and OpenAPI are reconciled. Phase 2 remains fully visible and governed, enabling future expansion without destabilizing the trust and consent foundations.

14.2 Source References

- NexHire Part 1 - Scope, Evidence and Solution v2.0
- NexHire Part 2 - Requirements, Journeys and UX v2.0
- NexHire Part 3 - Production Architecture v2.0
- NexHire Part 4 - Second Brain Implementation Playbook v2.0
- NexHire Team ALPHA Validated Prototype v2.0
- NexHire Master Project Documentation and Team Handover Guide

Appendix A - Requirement Identification Convention

BR identifies a business rule, FR a functional requirement, NFR a non-functional requirement, UC a use case, M a module, ADR an architecture decision record, PDR a product decision record, API an interface contract and TC a test case. Identifiers are never reused after publication.

14.3 v3.1 Amendment - Company Verification as a Complete Trust Workflow

Area	v3.1 Approved Rule
Business decision	Company verification is a mandatory trust gate for public vacancy publishing. A company may create and save Draft vacancies while unverified, but it cannot activate or publicly publish them until the verification state is Verified.
Employer outcome	The employer receives a guided verification workspace, evidence checklist, status tracking, administrator feedback, correction/resubmission path, and a clear publishing eligibility outcome.
Administrator outcome	Administrators receive a review queue, evidence detail, risk cues, history, decision reason capture, and auditable Approve, Reject, and Request More Information actions.
Candidate outcome	Candidates see the Trust Verified badge only for employers with a current approved verification record. The badge is not a match-score factor and does not imply endorsement of a specific vacancy.
Governance	Verification may expire or be suspended. Re-verification is required when legal identity, ownership, risk, or evidence validity materially changes.

This amendment is authoritative for Company Verification. Where an earlier summary is less detailed, this amendment and Volume 05 provide the complete implementation and UX contract.